# ATTS: Agentic Test-Time Scaling via Adaptive Exploration

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## Abstract

Test-time scaling has become a key mechanism for improving large language model reasoning, but existing approaches often rely either on rigid heuristic allocation rules or on explicit fine-grained search over intermediate reasoning steps. We introduce ATTS, a delegated sequential test-time scaling framework in which a lightweight orchestrator adaptively controls a pool of candidate solutions through a single action: `explore`, which launches a fresh, independent solver. The orchestrator decides when to stop exploring and directly synthesizes the final answer from the accumulated evidence. The framework defines an extensible action space over solver dispatch decisions; we instantiate it with a single parameterless `explore` tool and show that even this minimal configuration achieves strong results. We evaluate ATTS on six benchmarks spanning mathematical reasoning, scientific question answering, code generation, and multimodal reasoning. By stopping early when evidence converges, ATTS achieves competitive or superior accuracy with substantially fewer API calls than fixed-budget alternatives: 84.77% on GPQA-Diamond, 82.29% on LiveCodeBench, 56.00% on HLE-Verified, and 23.20% on BabyVision. An ablation replacing the orchestrator's direct synthesis with a separate integrator call degrades or fails to improve accuracy on three of four benchmarks, indicating that stateful evidence accumulation during exploration is critical to the method's effectiveness.

## 1 Introduction

The allocation of inference-time compute has emerged as a central frontier for improving the reasoning capabilities of large language models. Recent work on test-time compute is commonly organized around sequential scaling, parallel scaling, and hybrid search. The compute-allocation perspective of Snell et al. [35] demonstrates that no single strategy is uniformly optimal across prompts of varying difficulty. At the same time, simply allocating more compute is not enough: naive scaling can saturate, and in sampling-based settings can even degrade through poor coverage and reward hacking [14]. The central question is therefore not whether to scale test-time compute, but how to allocate it adaptively and efficiently.

Existing approaches often answer this question with either heuristics or explicit micro-level search. Predictive routing methods such as DiffAdapt use lightweight difficulty estimates to choose an inference regime before generation begins. However, these decisions remain open-loop and cannot react to evidence that appears during reasoning [22]. At the other extreme, Tree of Thoughts and process-reward frameworks such as Math-Shepherd perform closed-loop adaptation through explicit tree expansion, pruning, and step-level verification [44, 39]. Recent agentic test-time scaling methods add another variant of this pattern, but many still rely on statistical control rules such as vote entropy, confidence-triggered halting, or parallel mixtures of tool-using agents [19, 9, 20]. An underexplored alternative is to delegate compute allocation to an LLM orchestrator that performs closed-loop

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sequential decisions over a pool of candidate hypotheses, without explicit search trees or handcrafted stopping thresholds.

We introduce ATTS, a delegated sequential test-time scaling framework that separates *control* from *solution generation*. A lightweight orchestrator manages a growing pool of candidate solutions through a single tool, `explore`, which dispatches a fresh, independent solver on the original problem. When the orchestrator judges the pool to be sufficiently informative, it stops exploring and synthesizes the final answer directly from the accumulated evidence. ATTS treats compute allocation as closed-loop decision-making over an extensible action space: each `explore` call can in principle specify which solver to invoke, at what reasoning effort, and with what decomposition strategy. We evaluate the simplest instantiation—a single parameterless tool whose only decision is when to stop—and show that even this minimal agent achieves competitive accuracy at substantially lower cost. A multi-model variant that adds model selection to the action space yields further gains, supporting the design principle that the orchestrator's value scales with the richness of its decision space.

We evaluate ATTS across six benchmarks spanning mathematical reasoning, scientific question answering, code generation, and multimodal reasoning: HLE-Verified [45], LiveCodeBench [18], BabyVision [7], AIME 2025 and AIME 2026 [27, 28], and GPQA-Diamond [31]. Because the orchestrator is benchmark-agnostic, the same policy transfers across text-only and image-conditioned tasks.

## 2 Related Work

Recent work on test-time compute for large language models is commonly discussed under three broad families: sequential scaling, parallel scaling, and hybrid search. Following the compute-allocation perspective of Snell et al. [35], we use this taxonomy only to position ATTS relative to existing inference-time strategies.

**Sequential Scaling.** Sequential scaling increases reasoning depth by allocating additional computation along a single evolving trajectory. Representative examples include self-correction methods such as Self-Refine [25] and budget-controlled decoding such as s1 with Budget Forcing [29]. A critical negative result in this family is that LLMs cannot self-correct reasoning through prompting alone—post-correction accuracy typically drops [15]. This motivates architectures like ATTS that separate the evaluation role (orchestrator) from the solution role (explorer), rather than asking the same model to fix its own errors. At a larger scale, long-thinking models such as DeepSeek-R1 [13] internalize sequential scaling through reinforcement learning, spontaneously developing self-verification and extended chain-of-thought. ATTS pursues the complementary approach of agentic orchestration at inference time, requiring no additional training.

**Parallel Scaling.** Parallel scaling increases reasoning width by drawing multiple independent candidates and then aggregating or reranking them. The classical instance is self-consistency, which samples diverse reasoning paths and marginalizes over their final answers [41]. Brown et al. [4] show that coverage scales log-linearly with the number of repeated samples, but the resulting selection and verification bottleneck grows correspondingly—a gap that motivates ATTS's sequential evidence-management approach. Universal Self-Consistency extends aggregation to free-form answers by using the LLM itself to select among candidates [8]; ATTS's synthesis step can be seen as a sequential, evidence-accumulating version of this idea. More recent work pairs this width expansion with learned reward models or verifier models that score full responses, enabling reward-model-based selection rather than simple majority voting. This line includes outcome-reward reranking in the compute-scaling analysis of Snell et al. [35] and modern open reward models designed explicitly for majority voting inference, such as Skywork-Reward-V2 [21]. Recent theoretical work further shows that naive majority voting scaling is not uniformly monotonic: its effectiveness depends on policy coverage over good responses, and overly aggressive scaling can degrade performance through reward hacking [14]. In a complementary line of work, Chen et al. [6] argue that this issue is not only an inference-time phenomenon but also a training-time one, showing that standard cross-entropy fine-tuning can become misaligned with pass@$N$ scaling by making models overconfident. RASC introduces adaptive early stopping by evaluating the quality of each reasoning path, reducing sample usage while maintaining accuracy [38]. In multimodal settings, the same pattern appears in VisualPRM, which scores candidate multimodal reasoning traces for reranking [40]. The common structure is that

candidates are generated independently and the additional compute is spent on breadth plus post hoc selection.

**Hybrid Search.** Hybrid methods combine depth and width by repeatedly expanding multiple partial trajectories and pruning them with search or reward signals. Tree of Thoughts is an early and influential example: it generalizes chain-of-thought into a tree over intermediate "thoughts," allowing lookahead, self-evaluation, and backtracking [44]. Subsequent work strengthens this paradigm with more explicit search and value estimation, for example by integrating Monte Carlo Tree Search with iterative preference learning [43] or by introducing step-level process reward models such as Math-Shepherd to provide dense supervision over intermediate reasoning steps [39]. More recent hybrid systems such as Recursive Self-Aggregation explicitly combine population-level candidate aggregation with iterative refinement, showing that depth and width can be interleaved rather than treated as disjoint scaling axes [37]. These methods occupy the middle ground between pure sequential refinement and pure parallel sampling. We position ATTS against this landscape as a delegated sequential orchestrator rather than an explicit tree-search method.

**Agentic Test-Time Scaling.** Recent work studies test-time scaling explicitly for agents. CATTS allocates compute using vote-derived uncertainty statistics [19]; TUMIX scales through parallel heterogeneous agents with confidence-based early stopping [9]; and General AgentBench benchmarks sequential and parallel scaling for open-ended LLM agents [20]. On the adaptive stopping side, Manvi et al. [26] show that LLMs can predict mid-generation whether restarting will yield a better response. Bilal et al. [2] propose a PRM-guided adaptive framework. AB-MCTS dynamically decides whether to expand width (new candidates) or depth (refine existing ones) via adaptive branching [17]. These works predominantly rely on statistical routing rules, parallel mixtures, or explicit reward signals. A separate line of work on multi-agent debate [10, 42] shows that symmetric multi-round deliberation induces a martingale over belief trajectories, providing no expected accuracy improvement beyond majority voting. ATTS differs from debate systems in a critical structural way: candidates are generated independently (no inter-agent influence), and only the orchestrator accumulates evidence—an asymmetric architecture that avoids the martingale limitation. This design is closer to the input-adaptive compute allocation of Damani et al. [11], who learn a difficulty predictor to allocate budget across instances, but ATTS uses the LLM itself as the allocator rather than a trained predictor.

# 3 Methodology

## 3.1 Problem Formulation

Given an input problem $x$, the goal is to produce a final answer under a bounded test-time budget $T$. ATTS maintains a candidate pool $\mathcal{C}_t = \{c_1, \ldots, c_n\}$, where each candidate contains an answer, free-form reasoning, an approach summary, and an optional self-reported confidence score.

## 3.2 The ATTS Loop

The method proceeds as follows. A lightweight orchestrator repeatedly decides whether to call `explore` or to stop and synthesize. It never solves the problem itself.

**Explore.** Each `explore` call dispatches a fresh solver on the original problem $x$. The solver works from scratch—it does not see or continue previous candidates—and returns a structured candidate that is appended to $\mathcal{C}_t$. This fresh-start design encourages diversity: different strategies, error modes, and confidence profiles emerge independently across candidates.

**Stopping.** The orchestrator can stop at any point before the exploration budget $T$ is exhausted. The stopping decision is made by the orchestrator model itself based on its assessment of the candidate pool; no external stopping criterion or confidence threshold is imposed. The prompt encodes principles for this judgment: genuine convergence means independent solvers arriving at the same answer through different methods, repeated failures indicate the problem exceeds solver capability, and each additional explore call has a cost that must be justified by expected information gain. If

**Algorithm 1** ATTS: Adaptive Test-Time Scaling

**Require:** Problem $x$, exploration budget $T$, orchestrator LLM $\mathcal{O}$, solver LLM $\mathcal{S}$
**Ensure:** Final answer $y$

```
C ← ∅ {candidate pool}
for t = 1 to T do
    action ← O(x, C) {decide: explore or stop}
    if action = stop then
        break
    end if
    c_t ← S(x) {fresh, independent solve}
    C ← C ∪ {c_t}
end for
y ← O(x, C) {synthesize final answer}
return y
```

the budget is exhausted without an explicit stop, the method falls back to the highest-confidence candidate in the pool.

**Synthesis.** When the orchestrator stops, it synthesizes the final answer directly from $\mathcal{C}_t$. Because it has observed the full multi-turn history—each candidate's arrival, its reasoning, and its own deliberation about whether to continue—it has richer context than a separate aggregator that only sees the final pool. The orchestrator compares candidates, reconciles partial evidence, and produces a final answer that reflects the aggregate evidence.

Algorithm 1 summarizes the procedure. The control logic is benchmark-agnostic: only the explorer prompt and answer schema change across tasks. The orchestrator prompt encodes its decision logic as declarative principles (e.g. "a single candidate does not constitute sufficient evidence") rather than prescriptive rules, paired with worked examples that demonstrate how these principles apply in concrete scenarios. Appendix A gives the full implementation details including representative prompts.

### 3.3 Rational Metareasoning Perspective

The ATTS orchestrator can be understood through the lens of *rational metareasoning* [32]: rather than solving the problem directly, it allocates computational resources (explore calls) to maximize expected answer quality under a budget constraint. De Sabbata et al. [12] operationalize this framework by training LLMs with a VOC-derived reward that penalizes unnecessary reasoning steps; ATTS pursues the same principle at a coarser granularity, treating each `explore` call as the unit of computation whose marginal value is assessed. Each explore result updates the orchestrator's *belief state*—its accumulated evidence about which answer is correct—and the next action is chosen based on the current belief. This sequential, memory-dependent structure distinguishes ATTS from memoryless strategies such as Best-of-$N$, where all candidates are generated independently and no intermediate evidence informs subsequent sampling.

From this perspective, the orchestrator performs *budget-constrained optimization in belief space*. Each explore reduces uncertainty about the correct answer by adding an independent evidence sample, and the choice of action configuration (which model, at what cost) determines the expected information gain. The stopping criterion—whether additional exploration has positive marginal value—is the natural analogue of a convergence check. This framing yields a concrete design principle: the value of the ATTS orchestrator is proportional to the richness of its action space. When the only action is "explore once more with the same model," the orchestrator can only decide *when* to stop. With heterogeneous models, it also decides *what* to compute. As the action space expands further (e.g. solver effort level, time budget, decomposition strategy), the orchestrator gains additional degrees of freedom, enabling finer-grained adaptation to problem difficulty.

### 3.4 Multi-Model Extension

We extend ATTS to multi-model deployment by adding model selection to the orchestrator's decision space. The orchestrator (Claude Sonnet 4.6) dispatches explores to a pool of three models—Haiku (cheapest), Sonnet (mid-range), and Opus (strongest)—each with a per-question budget limit. Model profiles inform dispatch decisions: the orchestrator starts with cheaper models, escalates on disagreement or repeated failure, and treats cross-model agreement as stronger evidence than same-model convergence. This enables cost-aware exploration strategies where easy problems can be solved entirely with cheap models while hard problems automatically escalate to more capable solvers. An *exploration effort* parameter controls stopping aggressiveness: Low effort permits early stopping when any two candidates converge, Medium effort requires at least three explores across at least two different models, and High effort requires at least five explores using all three models.

## 4 Experiments

We evaluate on six benchmarks: HLE-Verified [45], LiveCodeBench [18], BabyVision [7], AIME 2025 and AIME 2026 [27, 28], and GPQA-Diamond [31]. The main text presents results on HLE-Verified, LiveCodeBench, BabyVision, and GPQA-Diamond, which collectively span mathematical reasoning, scientific question answering, code generation, and multimodal reasoning. AIME 2025 and AIME 2026 are deferred to Appendix B because the backbone model already achieves near-ceiling accuracy on these 30-problem benchmarks, limiting the statistical resolution for method comparison. Appendix F collects the full official task definitions, grading rules, source repositories, and leaderboard alignment notes for all benchmarks.

### 4.1 Experimental Setup

Across all experiments, the underlying backbone model is Claude Sonnet 4.6 (`claude-sonnet-4-6`), used for the orchestrator, explorer, and all baselines unless otherwise noted. Claude Haiku, Sonnet, and Opus are three capability tiers in the Claude model family, ordered from smallest/cheapest to largest/most capable. All generation uses the model's default sampling parameters (no explicit temperature or top-$p$ override) with extended thinking enabled at a budget of 32,000 tokens. Extended thinking allows the model to allocate a portion of its output budget to internal chain-of-thought reasoning before producing the final structured response; all baselines use the same setting for fair comparison. The exploration budget is $T{=}8$ `explore` calls per question for single-model runs and $T{=}16$ for multi-model runs (per-model limits: Haiku 8, Sonnet 8, Opus 4).

We compare against six baselines, all sharing the same backbone model and answer interface so that only the inference-time strategy differs (full details in Appendix A): Pass@1 (single rollout), Majority Voting ($N{=}8$ independent candidates, most-frequent answer; reported only on benchmarks where direct answer aggregation is well-defined), Self-Refine [25] (iterative feedback-revision on a single draft), Budget Forcing [29] ($N$ sequential rounds with forced continuation via "Wait"), reward-model reranking using Skywork-Reward-V2 [21, 34] for text benchmarks or VisualPRM [40, 30] for multimodal benchmarks, and LLM Selection ($N{=}8$), which feeds all pre-cached candidates to the backbone model in a single call and asks it to analyze, compare, and synthesize a final answer. LLM Selection isolates the contribution of LLM-based aggregation from adaptive sequential exploration.

For ATTS, the controller issues up to the exploration budget of `explore` calls before stopping and synthesizing the final answer directly, as described in Section 3. The comparison across all rows isolates differences in test-time inference strategy rather than differences in training data, parameter count, or post-training recipe.

**Cost accounting.** All costs reported as \$/q reflect API token charges only, computed from per-model pricing and recorded token usage. For ATTS, this includes orchestrator tokens, explorer tokens, and (when applicable) integrator tokens. For reward-model reranking, the reported cost covers the $N{=}8$ candidate generation calls; the reward model itself runs locally and incurs no API cost but does require GPU compute not reflected in the dollar figure. Grading costs (LLM judge calls for HLE and BabyVision) are excluded from all rows. In all tables, **bold** marks the best accuracy per group and the lowest \$/q among test-time scaling methods.

Table 1: Main results on four benchmarks. Top left: HLE-Verified (Accuracy). Top right: LiveCodeBench (Pass@1, overall and by difficulty). Bottom left: GPQA-Diamond (Accuracy). Bottom right: BabyVision (Avg@3, overall and by task family). Each subtable shows zero-shot model references (upper) and test-time scaling methods (lower). **Bold** marks the best per group; lowest $/q among test-time methods is also bolded.

*(a) HLE-Verified*

| | Acc. (%) | $/q |
|---|---|---|
| *Zero-Shot Models* | | |
| Gemini 3 Pro | 48.93 | – |
| GPT-5.2-High | **52.48** | – |
| Claude Opus 4.5 | 48.16 | – |
| Grok 4.1 Fast Reasoning | 43.07 | – |
| Claude Opus 4.6 | 50.16 | – |
| DeepSeek-V3.2 | 47.45 | – |
| Qwen3-Max-Thinking | 48.48 | – |
| *Test-Time Scaling Methods* | | |
| Pass@1 | 48.00 | 0.50 |
| Self-Refine | 53.00 | **0.41** |
| Budget Forcing | 51.00 | 0.60 |
| Skywork-Reward-V2 | 52.00 | 4.41 |
| LLM Selection ($N$=8) | **58.00** | 4.53 |
| ATTS (Ours) | 56.00 | 1.90 |
| ATTS Multi-model Med | 57.00 | 1.63 |

*(b) LiveCodeBench*

| | Pass@1 | Easy | Med. | Hard | $/q |
|---|---|---|---|---|---|
| *Zero-Shot Models* | | | | | |
| O4-Mini (High) | **80.2** | **99.1** | **89.4** | **63.5** | – |
| O3 (High) | 75.8 | **99.1** | 84.4 | 57.1 | – |
| O4-Mini (Med.) | 74.2 | 98.2 | 86.5 | 52.7 | – |
| Gemini-2.5-Pro | 73.6 | **99.1** | 87.2 | 50.2 | – |
| DeepSeek-R1 | 73.1 | 98.7 | 85.2 | 50.7 | – |
| *Test-Time Scaling Methods* | | | | | |
| Pass@1 | 77.14 | 95.35 | 88.46 | 60.00 | **0.24** |
| Self-Refine | 80.57 | **100.00** | 90.38 | 63.75 | 0.32 |
| Budget Forcing | 80.00 | **100.00** | 88.46 | 63.75 | 0.50 |
| Skywork-Reward-V2 | 78.86 | **100.00** | 90.38 | 60.00 | 1.41 |
| LLM Selection ($N$=8) | 81.14 | **100.00** | 86.54 | 67.50 | 1.51 |
| ATTS (Ours) | 82.29 | **100.00** | 90.38 | 67.50 | 0.53 |
| ATTS Multi-model Med | **85.63** | **100.00** | **92.31** | **73.42** | 0.69 |

*(c) GPQA-Diamond*

| | Acc. (%) | $/q |
|---|---|---|
| *Zero-Shot Models* | | |
| Gemini 3.1 Pro Preview | **94.1** | – |
| GPT-5.4 (xhigh) | 92.0 | – |
| GPT-5.3 Codex (xhigh) | 91.5 | – |
| Gemini 3 Pro Preview (high) | 90.8 | – |
| GPT-5.2 (xhigh) | 90.3 | – |
| *Test-Time Scaling Methods* | | |
| Pass@1 | 74.24 | **0.13** |
| Majority Voting | 73.74 | 0.90 |
| Self-Refine | 73.10 | 0.23 |
| Budget Forcing | 77.16 | 0.42 |
| Skywork-Reward-V2 | 76.14 | 0.90 |
| LLM Selection ($N$=8) | 77.16 | 0.97 |
| ATTS (Ours) | **84.77** | 0.36 |
| ATTS Multi-model Med | 81.82 | 0.39 |

*(d) BabyVision*

| | Overall | Fine-gr. | Tracking | Spatial | Pattern | $/q |
|---|---|---|---|---|---|---|
| Human | 94.1 | 92.3 | 94.6 | 94.7 | 97.8 | – |
| *Zero-Shot Models* | | | | | | |
| Gemini3-Pro-Preview | **49.7** | **46.2** | **43.4** | **53.7** | 53.9 | – |
| GPT-5.2 | 34.4 | 27.3 | 34.9 | 35.2 | **54.9** | – |
| Doubao-1.8 | 30.2 | 39.2 | 15.7 | 24.7 | 37.7 | – |
| Qwen3-VL-Plus | 19.2 | 21.8 | 11.5 | 18.1 | 25.5 | – |
| Grok-4 | 16.2 | 11.0 | 24.1 | 13.2 | 24.8 | – |
| *Test-Time Scaling Methods* | | | | | | |
| Pass@1 | 19.59 | 20.86 | 19.28 | 19.78 | 15.69 | **0.07** |
| Majority Voting | **23.20** | **23.93** | 21.69 | 23.08 | **23.53** | 0.50 |
| Self-Refine | 21.39 | 20.25 | 22.89 | 20.88 | **23.53** | 0.42 |
| Budget Forcing | 21.91 | 20.86 | **28.92** | 21.98 | 13.73 | 0.52 |
| VisualPRM | 20.88 | 20.86 | 21.69 | 21.98 | 17.65 | 0.50 |
| LLM Selection ($N$=8) | **23.20** | 23.31 | 25.30 | 23.08 | 19.61 | 0.56 |
| ATTS (Ours) | **23.20** | 22.09 | 24.10 | **26.37** | 19.61 | 0.27 |
| ATTS Multi-model Med | 22.42 | 22.09 | 27.71 | 21.98 | 15.69 | 0.43 |

**Run-to-run stability.** To assess variance, we run ATTS five times on GPQA-Diamond with different random seeds (varying the order of questions and orchestrator sampling). The results are highly stable: $80.20\% \pm 0.55\%$ (range: $79.8\%$–$80.8\%$, $\sigma = 0.55$), with cost $\$0.32 \pm \$0.01$/q. The standard deviation of 0.55 percentage points corresponds to a difference of approximately two questions across runs, indicating that the orchestrator's stopping and synthesis decisions are robust to sampling stochasticity. All other results in the paper are single-run evaluations; the GPQA stability analysis suggests that the reported numbers are representative.

### 4.2 Main Results

Table 1 presents the main results across all four benchmarks. For each benchmark, the upper group reproduces published zero-shot references for context, and the lower group reports test-time scaling methods evaluated under our unified setup. Zero-shot references are cited from the respective benchmark papers and official leaderboards; they use different model configurations and are therefore not directly comparable to our test-time scaling rows. HLE-Verified zero-shot rows report Revised-subset Acc [45] while our rows evaluate a 100-sample Gold subset. LiveCodeBench references follow the public generation leaderboard [18]. BabyVision references are from Chen et al. [7]. GPQA-Diamond references are from Rein et al. [31] and official records. Full benchmark details including dataset releases, grading protocols, and subset definitions are in Appendix F.

**HLE-Verified.** On HLE-Verified, ATTS achieves 56.00% accuracy at $1.90/q, outperforming Self-Refine (53.00%) and Budget Forcing (51.00%). LLM Selection reaches 58.00% but at $4.53/q—a $2.4\times$ cost increase for a 2-point gain—illustrating a cost-accuracy tradeoff where single-pass aggregation can achieve higher accuracy when cost is unconstrained, but adaptive exploration provides a more efficient operating point. The multi-model variant improves to 57.00% at $1.63/q, narrowing the accuracy gap while maintaining cost efficiency.

**LiveCodeBench.** For LiveCodeBench, the single-model ATTS configuration reaches 82.29% Pass@1 at $0.53/q, improving over Self-Refine (80.57%) and Budget Forcing (80.00%). The gains concentrate on Hard problems: 67.50% vs. 63.75% for both Self-Refine and Budget Forcing, while Easy (100%) and Medium (90.38%) accuracies are at or near ceiling. The multi-model variant further improves to 85.63% overall and 73.42% on Hard.

**GPQA-Diamond.** On GPQA-Diamond, ATTS achieves the highest accuracy among all test-time scaling methods (84.77%) at $0.36/q, substantially outperforming Budget Forcing (77.16%) and Skywork-Reward-V2 (76.14%). Self-Refine (73.10%) falls below Pass@1 (74.24%), consistent with the observation that its feedback model frequently accepts the initial draft without revision, and occasionally introduces errors when it does revise.

**BabyVision.** Among test-time scaling methods, ATTS ties with Majority Voting and LLM Selection at 23.20% overall accuracy on BabyVision, but at $0.27/q—roughly half the cost of either alternative. The largest per-subset margin appears on Spatial Perception, where ATTS reaches 26.37%.

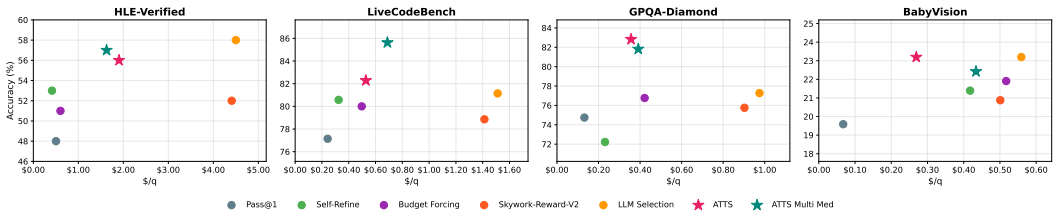


Figure 1: Cost-accuracy tradeoffs across the four main benchmarks (left to right: HLE-Verified, LiveCodeBench, GPQA-Diamond, BabyVision). Star markers denote ATTS variants; circles denote baselines. ATTS achieves competitive or superior accuracy at lower cost across all benchmarks.

**Overall trends.** Across benchmarks, ATTS is intended to occupy a middle ground between fixed-budget sequential refinement and purely parallel sampling. Direct one-shot generation is often already strong on easier exact-match tasks, but it can leave performance on the table once problems require either broader hypothesis coverage or evidence reconciliation across partially correct candidates. Majority Voting addresses the first issue by increasing candidate diversity, yet it still reduces the final decision to selection. The role of ATTS is to use the same test-time budget more adaptively: explore when the current pool is incomplete or inconsistent, and synthesize the final answer once the candidate set contains enough evidence.

This framing is especially useful on benchmarks where independently generated candidates can expose complementary reasoning fragments rather than merely redundant answers. On mathematical and scientific reasoning tasks, the expected gain comes from avoiding premature commitment to a single trajectory while still permitting a model-based final consolidation step. On code and multimodal reasoning tasks, the same controller provides a unified way to trade off breadth against synthesis without changing the benchmark-specific grading pipeline. Taken together, the results support our main claim: even a purposefully constrained single-tool interface can be sufficient to turn test-time scaling into a semantic compute-allocation problem rather than a fixed sampling schedule.

### 4.3 Ablation Studies

**Exploration effort.** The multi-model variant exposes an *exploration effort* parameter that controls stopping aggressiveness. Figure 2 shows the cost-accuracy tradeoff at three effort levels (Low,

Medium, High) across all four benchmarks, holding the model pool and budget fixed; the raw numbers are in Appendix C, Table 5. The Low-to-Medium transition yields the largest accuracy gain on every benchmark (up to +13 points on HLE, +8.6 on LiveCodeBench). Medium-to-High provides diminishing returns: on LiveCodeBench and BabyVision accuracy is flat while cost nearly doubles; only on HLE-Verified does High effort produce a meaningful further improvement (+3 points). Medium effort therefore represents the best cost-efficiency operating point across the benchmark suite.

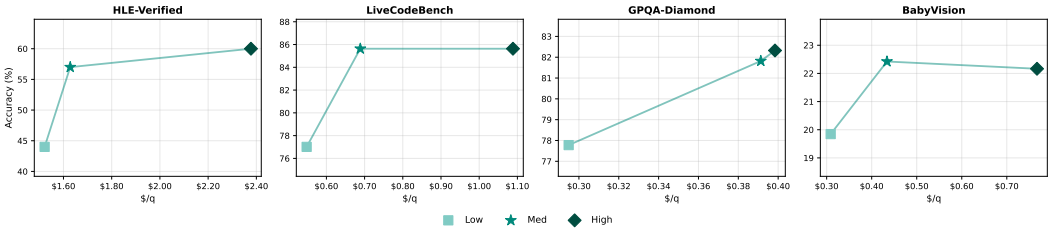


Figure 2: Effect of exploration effort on multi-model ATTS. Each panel shows the cost-accuracy tradeoff for Low, Medium, and High effort on one benchmark. The steep Low-to-Medium gain and flat Medium-to-High plateau are consistent across all benchmarks.

**Adding a separate integrator.** In the default ATTS architecture, the orchestrator produces the final answer directly from its accumulated evidence after deciding to stop exploring. We also evaluate a variant that adds a dedicated `integrate` step: a separate model call that receives all candidates and synthesizes the final answer independently.

Table 2: Effect of adding a separate integrator. The default ATTS architecture has the orchestrator synthesize directly; the "+ Integrator" variant delegates synthesis to a separate model call.

| | Default (no integrator) | | + Integrator | |
|---|---|---|---|---|
| | Acc. (%) | $/q | Acc. (%) | $/q |
| HLE-Verified | 56.00 | 1.90 | 57.00 | 2.66 |
| LiveCodeBench | 82.29 | 0.53 | 80.57 | 0.72 |
| BabyVision | 23.20 | 0.27 | 23.20 | 0.38 |
| GPQA-Diamond | 84.77 | 0.36 | 76.14 | 0.44 |

Across all four benchmarks, adding a separate integrator does not consistently improve accuracy and always increases cost (Table 2). On GPQA-Diamond, accuracy drops from 84.77% to 76.14%; on LiveCodeBench, from 82.29% to 80.57%. On HLE-Verified, the integrator variant gains 1 point (57.00% vs. 56.00%) but at substantially higher cost ($2.66/q vs. $1.90/q). The degradation likely stems from the integrator having less context for synthesis: it only sees the final candidate pool in a single-turn call, whereas the orchestrator has observed the full multi-turn conversation including each candidate's arrival and the reasoning behind its own exploration decisions.

**Orchestrator model selection.** We ablate the orchestrator model while keeping the explorer fixed at Sonnet 4.6 (Table 6, Figure 3). The results show no monotonic relationship between model strength and orchestrator effectiveness: Opus achieves the highest accuracy on HLE (59%) and LiveCodeBench (83.4%), but drops to 77.16% on GPQA—well below Sonnet's 84.77%. Haiku performs comparably to Sonnet on HLE (57% vs. 56%).

These differences are not explained by stopping behavior. All three orchestrators use similar numbers of explores on each benchmark (e.g. 2.15–2.43 on GPQA) and read from the same candidate cache. The gap is therefore attributable to synthesis quality: given similar evidence, the orchestrators produce different final answers. The pattern is benchmark-dependent rather than capability-monotonic—the strongest model is the best orchestrator on some benchmarks but the worst on others. This suggests that the orchestrator role involves a task-specific interaction between the model's reasoning style and

the domain characteristics, rather than a simple function of model capability. Identifying the precise mechanism underlying this interaction remains an open question.

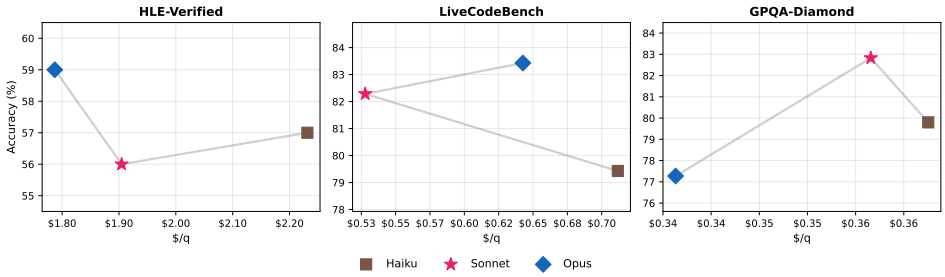


Figure 3: Effect of orchestrator model on ATTS accuracy and cost. Haiku is cheapest per call but over-explores, increasing total cost. Opus improves accuracy on HLE and LiveCodeBench but not on GPQA.

**Backbone model family.** To evaluate whether ATTS generalizes beyond a single model family, we run the full framework using GPT-5.2 (OpenAI) as the backbone for all three roles—orchestrator, explorer, and synthesizer—keeping the same exploration budget ($T$=8) and prompts as the Claude Sonnet configuration. We evaluate on two benchmarks (HLE-Verified and GPQA-Diamond) and test two reasoning effort levels to assess the effect of orchestrator reasoning depth.

Table 3: Backbone model ablation. All configurations use the default ATTS architecture (no separate integrator) with identical prompts and exploration budget. Reasoning effort controls the orchestrator's internal chain-of-thought budget. ATTS gain = ATTS accuracy − Pass@1 accuracy.

| Backbone | Effort | Benchmark | Pass@1 | ATTS | Gain | $/q |
|---|---|---|---|---|---|---|
| Claude Sonnet 4.6 | default | HLE | 48.00 | 56.00 | +8.0 | 1.90 |
| Claude Sonnet 4.6 | default | GPQA | 74.24 | 84.77 | +10.5 | 0.36 |
| GPT-5.2 | high | HLE | 55.00 | 57.00 | +2.0 | 0.26 |
| GPT-5.2 | high | GPQA | 56.06 | **71.07** | **+15.0** | **0.16** |

ATTS produces positive gains on both benchmarks for both model families (Table 3). On GPQA-Diamond, GPT-5.2 shows the largest absolute gain: +15.0 points over Pass@1 (56.06% → 71.07%), compared to Claude Sonnet's +10.5 points. The lower GPT-5.2 Pass@1 (56.06% vs. the 90.3% reported for GPT-5.2 in the GPQA leaderboard) reflects the use of `effort=low` for exploration, which limits reasoning tokens; the orchestrator operates at `effort=high` during synthesis. On HLE-Verified, the gain is smaller (+2.0 points) but still positive, and the cost is 7.3× lower than Claude. These results demonstrate that ATTS's adaptive exploration framework is not specific to any model family: the same prompts and control logic transfer to a different provider with consistent gains at substantially lower cost.

## 5 Limitations

ATTS inherits several limitations from the underlying language model and from its own orchestrator design. First, the method can only synthesize evidence that is already present in the candidate pool, so if all explored candidates share the same misconception, the orchestrator has little opportunity to recover. Second, the current framework does not provide formal guarantees for stopping optimality, calibration, or budget efficiency; the orchestrator is still a learned heuristic policy, even if its action space is intentionally simple. In practice, we observe that the orchestrator tends to converge early: on GPQA, 82% of questions use exactly two `explore` calls before stopping, even though the oracle best-of-$N$ curve continues to climb through $N$=8. This conservative exploration policy leaves headroom on the table, and learning a better stopping criterion remains an open problem. Third, because ATTS requires multiple model calls, it can increase latency and inference cost relative to direct generation on easy problems where extra exploration is unnecessary. Finally, the current evaluation focuses on

benchmarked reasoning tasks with fixed answer interfaces rather than on open-ended tool-use agents, so the present results should be interpreted as evidence about reasoning-oriented test-time scaling rather than as a complete account of agentic decision making.

## 6 Conclusion

The central empirical finding of this work is that an LLM orchestrator performing closed-loop decision-making over a minimal action space—a single `explore` tool—achieves competitive or superior accuracy to fixed-budget test-time scaling strategies while using substantially fewer API calls. A multi-model variant further improves to 57.00% and 85.63% on HLE-Verified and LiveCodeBench respectively. The ablation results sharpen this finding: removing the orchestrator's multi-turn context by delegating synthesis to a separate call degrades or fails to improve accuracy on three of four benchmarks, demonstrating that stateful evidence management—not just candidate diversity—drives the gains. For the broader field, these results suggest that test-time compute allocation is better approached as closed-loop decision-making by an LLM agent—with an extensible action space over solver dispatch—than as explicit tree search or fixed sampling. A key limitation is the orchestrator's conservative stopping behavior: on GPQA-Diamond, 82% of questions use exactly two `explore` calls, leaving substantial headroom relative to oracle best-of-$N$. Learning stopping policies that better exploit the full exploration budget remains an open problem. Code and cached experimental artifacts are available at `https://github.com/t2ance/ATTS`.

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# A Implementation Details

**Controller and backend setup.** ATTS is implemented as a delegated controller with two roles: an orchestrator and an explorer. The orchestrator never solves the task directly; it only decides whether to spend the remaining budget on another `explore` call or to stop and synthesize the final answer from the accumulated evidence. Each `explore` call returns a structured candidate containing an answer, supporting reasoning, an approach summary, and an optional confidence score. When the orchestrator decides to stop, it produces the final answer directly via the SDK's structured output mechanism. For the Claude implementation, we use the Claude Agent SDK [1] and expose `explore` as the only available delegated tool. We also evaluate a variant that adds a separate `integrate` tool that delegates final synthesis to a dedicated model call; see the ablation in Section 4.3. The same high-level controller is reused across benchmarks, while the underlying sub-model backend can be swapped without changing the controller logic.

**Prompt families.** ATTS uses two prompt families: a benchmark-agnostic orchestrator prompt and a benchmark-specific explorer prompt. The orchestrator prompt defines the controller role and explicitly instructs it not to solve the task itself. The explorer prompt asks for an independent solution from scratch under the target answer format. Representative prompt excerpts are shown in Listings 1 and 3. For Claude runs, these prompts are paired with structured-output schemas so that each sub-call returns machine-readable fields rather than free-form text only.

```
You are a meta-reasoning orchestrator. You manage a pool of candidate solutions for a
    problem.

After each explore, you will see the candidate result (answer, confidence, approach,
    reasoning).

Your final answer must be derived from candidate outputs. You may combine insights from
    multiple candidates, but you cannot introduce information that no candidate provided
    .

## Principles (HIGHEST PRIORITY)

1. You cannot solve problems yourself. Your only window into the problem is what solvers
    return. Reasoning about the problem content constitutes solving.
2. A single candidate, regardless of confidence, does not constitute sufficient evidence.
     Self-reported confidence is poorly calibrated.
3. Genuine convergence means independent solvers arriving at the same answer through
    different methods. Shared reasoning paths may reflect shared misconceptions.
4. Solver failure (timeout, empty answer) reflects problem difficulty. Repeated failures
    mean the problem is practically unsolvable.
5. CRITICAL: Each explore costs budget. Giving up is valid and preferred -- submit empty
    answer rather than wasting budget.

[6 worked examples follow, showing: starting exploration, disagreement resolution,
    timeout handling, escalation, and giving up.]
```

Listing 1: Orchestrator system prompt (single-model, explore-only variant). The same principles and examples are shared by the integrate and multi-model variants; only the finalize instruction differs.

```
You have access to multiple solver models of different capability and cost. Each explore
     call lets you choose which model to use.

## Model Profiles
Pricing (per million tokens): Haiku $1/$5, Sonnet $3/$15, Opus $5/$25.
[Public benchmark scores table: MATH-500, AIME, GPQA, HLE, ARC-AGI-2, SWE-bench, etc.]

## Adapted Principles
3. Candidates from different models that agree provide stronger evidence than same-model
     candidates.
4. A weaker model failing does not mean a stronger model will also fail. Start cheap,
     escalate on failure.

## Exploration Effort Constraints (appended to user message)
LOW:    Stop as soon as 2 candidates from any model converge.
MEDIUM: MUST call explore >= 3 times, MUST use >= 2 models.
HIGH:   MUST call explore >= 5 times, MUST use all 3 models.
```

Listing 2: Multi-model orchestrator prompt additions. The orchestrator receives model profiles with pricing and benchmark scores, and adapted principles for heterogeneous model selection.

```
Explorer:
You are an expert problem solver. Solve the given problem step by step.
If you cannot solve it exactly, give your best estimate and set confidence accordingly.
Schema: {approach, reasoning, answer, confidence}

Integrator:
You are an expert answer synthesizer. You will receive a problem and multiple candidate
     solutions.
Your job: (1) Analyze each candidate's reasoning individually, (2) Identify agreements
     and disagreements, (3) Judge which side has more rigorous reasoning, (4) Point out
     logical errors, (5) Produce a verified final answer.
Schema: {analysis, final_answer, reasoning}
```

Listing 3: Explorer prompt (left) and integrator prompt (right). Both share the same answer format rules enforced via structured output schema field descriptions.

**Schemas and benchmark-specific prompt instantiation.** Benchmark-specific adaptation is intentionally lightweight. Most answer-based tasks share the explorer schema `{approach, reasoning, answer, confidence}` and the integrator schema `{analysis, final_answer, reasoning}`. Only the answer interface changes across tasks. LiveCodeBench uses code-generation fields, AIME requires an integer final answer, GPQA expects one option letter, and multimodal benchmarks pass the associated image with the text question. This design keeps the controller fixed while preserving each benchmark's native answer format and grading protocol.

**Runtime behavior.** At runtime, the controller maintains the candidate pool and repeatedly calls `explore` until the exploration budget is exhausted or the controller decides to stop and synthesize. The same implementation supports both text-only and image-conditioned tasks, and all runs can optionally record the controller trace and sub-model outputs for later inspection. These implementation choices are meant to make the evaluation pipeline inspectable without changing the central adaptive exploration decision process.

**Grading.** LiveCodeBench and AIME use deterministic grading (code execution and exact integer matching, respectively). GPQA-Diamond uses exact option-letter matching, consistent with the official GPQA evaluation protocol [31]; as with any regex-based extraction, minor formatting variations in model output may introduce small amounts of grading noise. HLE-Verified and BabyVision use an LLM judge (Claude Haiku 4.5) that compares the predicted answer against the reference answer.

**Baseline implementations.** All baselines use the same backbone model and the same benchmark-specific answer format as ATTS, so the comparison isolates the inference-time strategy.

*Pass@1* generates a single independent response per question and returns its answer directly.

*Majority Voting* draws $N = 8$ independent responses and selects the most frequent answer.

*Self-Refine* [25] generates an initial draft, then enters a feedback-revision loop: a separate feedback call critiques the current draft and returns a correctness judgment; if the draft is judged incorrect, a revision call produces an updated draft conditioned on all prior drafts and feedbacks. The loop terminates when feedback returns a positive judgment or the iteration budget is exhausted. In practice, the feedback model frequently accepts the initial draft, so Self-Refine terminates after a single feedback call on the majority of questions, which explains its relatively low cost on most benchmarks.

*Budget Forcing* [29] runs $N$ sequential rounds on the same question. Round 1 generates an initial solution. Each subsequent round feeds the previous round's full reasoning trace back to the model with "Wait" appended, suppressing premature termination and forcing the model to continue deliberating before producing a new answer. All $N$ rounds run unconditionally, and the final round's answer is returned. Because continuation rounds extend the existing reasoning rather than solving from scratch, they tend to produce shorter completions and lower per-round cost.

*Reward-model reranking* (Skywork-Reward-V2 for text benchmarks, VisualPRM for multimodal benchmarks) draws the same $N = 8$ independent candidates, then scores each with a separately trained reward model and returns the highest-scoring answer. Skywork-Reward-V2 is an outcome reward model that assigns a single scalar score per response; VisualPRM is a process reward model that aggregates per-step scores. The reward models run locally and incur no API cost, but the candidate generation cost is identical to that of Majority Voting.

# B AIME Results

AIME 2025 and AIME 2026 are 30-problem final-answer mathematics benchmarks with exact integer-answer grading. Because the backbone model already achieves near-ceiling accuracy on these benchmarks (93.33% on AIME 2025 with a single attempt), the statistical resolution for method comparison is limited. We include them here for completeness.

Table 4 presents the AIME 2025 and AIME 2026 results side by side. Full benchmark details including the official answer-space constraint, source repositories, and leaderboard alignment are in Appendix, AIME 2025 and Appendix, AIME 2026.

Table 4: AIME results. Left: AIME 2025. Right: AIME 2026. Both are 30-problem final-answer benchmarks with exact integer-answer grading. Cited from official online records.

*(a) AIME 2025*

| | Acc. (%) | $/q |
|---|---|---|
| *Zero-Shot Models* | | |
| GPT-5.2 (high) | **100.00** | – |
| Step 3.5 Flash | 98.33 | – |
| Gemini 3 Flash | 97.50 | – |
| GLM 5 | 96.67 | – |
| DeepSeek-v3.2-Speciale | 95.83 | – |
| *Test-Time Scaling Methods* | | |
| Pass@1 | **93.33** | **0.21** |
| Majority Voting | **93.33** | 1.49 |
| Self-Refine | **93.33** | 3.13 |
| Budget Forcing | **93.33** | 0.52 |
| Skywork-Reward-V2 | 90.00 | 1.49 |
| ATTS (Ours) | **93.33** | 0.43 |

*(b) AIME 2026*

| | Acc. (%) | $/q |
|---|---|---|
| *Zero-Shot Models* | | |
| GPT-5.4 (xhigh) | **99.17** | – |
| GPT-5.2 (high) | 98.33 | – |
| Gemini 3.1 Pro Preview | 98.33 | – |
| Step 3.5 Flash | 96.67 | – |
| Claude-Opus-4.6 (High) | 96.67 | – |
| *Test-Time Scaling Methods* | | |
| Pass@1 | **96.67** | **0.22** |
| Majority Voting | 93.33 | 1.37 |
| Self-Refine | 90.00 | 0.23 |
| Budget Forcing | 93.33 | 0.60 |
| Skywork-Reward-V2 | **96.67** | 1.37 |
| ATTS (Ours) | **96.67** | 0.48 |

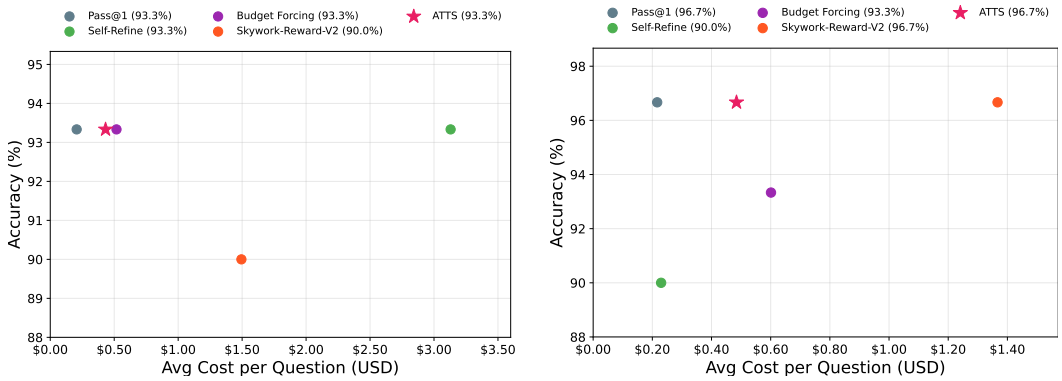


Figure 4: Cost-accuracy tradeoffs on AIME 2025 (left) and AIME 2026 (right).

**AIME 2025.** On AIME 2025, most test-time scaling methods converge to the same accuracy (93.33%), with only Skywork-Reward-V2 falling behind at 90.00%. The 30-problem dataset leaves little statistical room for differentiation, and the backbone model already solves 28 of 30 problems with a single attempt.

**AIME 2026.** On AIME 2026, Pass@1, ATTS, and Skywork-Reward-V2 all reach 96.67% (29/30), confirming near-ceiling performance. Self-Refine drops to 90.00%, likely because its feedback model misjudges some correct solutions and introduces errors during revision. As with AIME 2025, the small dataset size (30 problems) limits the granularity of method comparisons.

## C Ablation Data

Table 5: Effect of exploration effort on multi-model ATTS. Accuracy (%) and average cost per question ($/q) across benchmarks.

| | Low | | Medium | | High | |
|---|---|---|---|---|---|---|
| | Acc. | $/q | Acc. | $/q | Acc. | $/q |
| HLE-Verified | 44.00 | 1.52 | 57.00 | 1.63 | 60.00 | 2.38 |
| LiveCodeBench | 77.01 | 0.55 | 85.63 | 0.69 | 85.63 | 1.09 |
| BabyVision | 19.85 | 0.31 | 22.42 | 0.43 | 22.16 | 0.77 |
| GPQA-Diamond | 77.78 | 0.29 | 81.82 | 0.39 | 82.32 | 0.40 |

Table 6: Effect of orchestrator model on ATTS accuracy (%) and cost ($/q). Explorer is Sonnet 4.6 throughout.

| | Haiku orch | | Sonnet orch | | Opus orch | |
|---|---|---|---|---|---|---|
| | Acc. | $/q | Acc. | $/q | Acc. | $/q |
| HLE-Verified | 57.00 | 2.23 | 56.00 | 1.90 | 59.00 | 1.79 |
| LiveCodeBench | 79.43 | 0.71 | 82.29 | 0.53 | 83.43 | 0.64 |
| GPQA-Diamond | 80.20 | 0.36 | 84.77 | 0.36 | 77.16 | 0.34 |

## D Theoretical Analysis

The following results are a specialization of Blackwell's informativeness theorem [3] to deterministic information loss in the ATTS setting. Let $\mathcal{O}$ denote the full orchestrator observation space (the complete multi-turn history including each candidate's arrival, deliberation context, and stopping reasoning), $\mathcal{S}$ the integrator observation space (the final candidate pool), and $g : \mathcal{O} \to \mathcal{S}$ the

compression that discards arrival order and deliberation context. Let $\mathcal{A}$ be the answer space and $u : \mathcal{O} \times \mathcal{A} \to \mathbb{R}$ a utility function.

**Theorem 1** (Replication principle)**.** *For any integrator rule $d_I : \mathcal{S} \to \mathcal{A}$, there exists an orchestrator rule $d_O : \mathcal{O} \to \mathcal{A}$ with $u(\sigma, d_O(\sigma)) = u(\sigma, d_I(g(\sigma)))$ for all $\sigma \in \mathcal{O}$.*

*Proof.* Set $d_O = d_I \circ g$. □

The orchestrator can always replicate the integrator; the converse does not hold. When $g$ is non-injective, the integrator is provably handicapped:

**Theorem 2** (Information loss degrades synthesis)**.** *Suppose $g(\sigma_1) = g(\sigma_2)$ for distinct $\sigma_1, \sigma_2 \in \mathcal{O}$ with uniquely optimal answers $a_1^* \neq a_2^*$. Then every integrator rule $d_I$ is suboptimal on at least one:*

$$u(\sigma_1, d_I(g(\sigma_1))) < u(\sigma_1, a_1^*) \quad \textit{or} \quad u(\sigma_2, d_I(g(\sigma_2))) < u(\sigma_2, a_2^*).$$

*Proof.* Since $g(\sigma_1) = g(\sigma_2)$, $d_I$ outputs the same answer $a$ on both. If $a = a_1^*$ (optimal on $\sigma_1$), then $a \neq a_2^*$, and unique optimality on $\sigma_2$ gives $u(\sigma_2, a) < u(\sigma_2, a_2^*)$. Otherwise $u(\sigma_1, a) < u(\sigma_1, a_1^*)$. □

The integrator receives only $\mathcal{C}_\tau$ in a single call—a lossy summary of the orchestrator's full multi-turn history. Theorem 2 predicts that whenever the discarded context (reasoning approach diversity, convergence trajectory, stopping rationale) is decision-relevant, synthesis quality must degrade. The accuracy drops in Table 2—6.57 points on GPQA-Diamond, 1.72 on LiveCodeBench—are instances of this information-loss penalty. Formally verified proofs in Lean 4 are included in the code release.

# E Qualitative Analysis

To understand *how* ATTS improves over Pass@1, we examine the 22 GPQA-Diamond questions where ATTS (Sonnet orchestrator) produces a correct answer but the first exploration does not. Table 7 categorizes these by the orchestrator's mechanism of improvement.

Table 7: How ATTS improves over Pass@1 on GPQA-Diamond. *Majority correction*: multiple explores converge on the correct answer, overriding an incorrect first candidate. *Minority extraction*: the orchestrator identifies a correct minority candidate by evaluating reasoning quality. In one question, ATTS changes a correct Pass@1 answer to an incorrect one.

| Orchestrator behavior | Count | Fraction |
|---|---|---|
| Majority correction | 18 | 82% |
| Minority extraction | 4 | 18% |
| Total rescued (Pass@1 wrong → ATTS correct) | 22 | |
| Total hurt (Pass@1 correct → ATTS wrong) | 1 | |
| **Net gain** | **+21** | |

Most improvements (82%) arise from the diversity of independent explorations: when the first candidate is wrong, subsequent explorations are likely to find the correct answer, and the orchestrator follows the emerging consensus. The remaining 18% involve the orchestrator overriding the explore majority based on reasoning quality—a capability that goes beyond simple voting. Listings 4 and 5 show representative examples of each mechanism.

```
Question: A dicarboxylic acid containing a cis-alkene was dehydrated to
the corresponding anhydride [...] reacted with 1,2,3,4-tetramethyl-
cyclopentadiene [...] How many peaks are in the CHCH3 region?
Options: (A) 2 peaks  (B) 3 peaks  (C) 4 peaks  (D) 8 peaks
Gold: A

Explore #1: answer=D, confidence=0.85
  Approach: Identify anhydride, determine Diels-Alder stereochemistry.

Explore #2: answer=A, confidence=0.88
  Approach: Maleic anhydride + tetramethylcyclopentadiene gives endo
  bicyclo[2.2.1] product. The methyl groups split into two sets by
```

```
  symmetry: bridge methyls (6H singlet) and ring methyls (6H singlet).

Explore #3: answer=A, confidence=0.88
  [Independent confirmation via the same endo-selectivity argument.]

Orchestrator synthesis (after 5 explores, 3 favoring A):
  "Three independent explorations converged on A through consistent
   analysis of the endo product's symmetry. The dissenting candidates
   incorrectly assumed exo selectivity."
  Final answer: A, confidence=0.75. CORRECT.
```

Listing 4: Majority correction: a Diels-Alder/NMR problem (GPQA-Diamond). The first explore gives the wrong answer (D); subsequent explores converge on A through independent analyses. The orchestrator follows the consensus.

```
Question: Which of the following molecules has C3h symmetry?
(A) Triisopropyl borate  (B) Triphenylene trianhydride
(C) Benzene trianhydride  (D) Quinuclidine
Gold: B

Explore #1: answer=A, confidence=0.92
  "Triisopropyl borate: B is sp2 planar, C3 axis + sigma_h from the
   BO3 plane, no C2 or sigma_v. Point group: C3h."
Explore #2: answer=B, confidence=0.85
  "Triisopropyl borate: isopropyl groups adopt a propeller arrangement
   -- not planar, so no sigma_h. True point group: C3, not C3h."
Explore #3: answer=A, confidence=0.88
  [Repeats the sp2-planar argument for molecule A.]
Explore #4: answer=B, confidence=0.82
  [Identifies that benzene trianhydride (C) has 3 perpendicular C2
   axes -> D3h, while triphenylene trianhydride (B) lacks C2 due to
   inequivalent positions in the fused ring system -> C3h.]
[... 3 more explores: 2 favor A, 1 favors B. Final tally: A=4, B=3.]

Orchestrator synthesis (after 7 explores):
  "The candidates favoring A assume the isopropyl groups preserve
   sigma_h, but Explore #4 correctly notes that the propeller
   conformation breaks this plane. Triphenylene trianhydride (B) has
   C3h intrinsic to its rigid topology -- positions 1,2 in triphenylene
   are inequivalent, preventing C2 axes that would elevate to D3h."
  Final answer: B, confidence=0.62. CORRECT.
```

Listing 5: Minority extraction: a molecular symmetry problem (GPQA-Diamond). The explore majority (4/7) favors molecule A, but the orchestrator identifies a flaw in the majority's reasoning and selects the minority answer B.

**Emergent behavior: selective exploration bypass.** When using Opus 4.6 as orchestrator, we observe a distinct behavior absent in weaker models: the orchestrator occasionally bypasses exploration entirely, answering directly from its own knowledge. On GPQA-Diamond, Opus skips exploration on 30 of 198 questions (15%), achieving 90% accuracy on these self-answered questions—substantially higher than its 75% accuracy on explored questions. This suggests that stronger orchestrators develop an implicit sense of which problems are within their direct competence, selectively invoking the exploration mechanism only when needed.

# F Benchmark Details

**HLE-Verified.** We use the official HLE-Verified dataset release `skylenage/HLE-Verified` [33], which partitions the 2,500 audited HLE items into Gold (668), Revision (1,143), and Uncertain (689) subsets [45]. HLE-Verified preserves the original HLE answer-based evaluation semantics: each item contains a problem statement, a final answer, and sometimes a rationale, but official scoring is driven by the problem and final answer, with rationale used only as auxiliary evidence during verification and revision [5, 45]. In the HLE-Verified evaluation paper, the reported baselines are run on the text-only portion of HLE using the official HLE system prompt, each model's default recommended decoding configuration, and five independent rollouts per item; the paper reports both *avg5 Accuracy* and *Calibration Error*, with calibration computed by the official HLE evaluation code [45]. In this paper, HLE reporting is aligned to the Gold-subset setting, while Table 1(a) reproduces the official revised-subset accuracy references from Zhai et al. [45] for context. Our grading uses

an LLM-as-judge pipeline with Claude Haiku 4.5 as the judge model, which determines semantic equivalence between the predicted and gold answers.

**LiveCodeBench.** We use the LiveCodeBench code generation release `livecodebench/code_generation_lite` [23], configuration `v6` [18]. LiveCodeBench is a continuously updated programming benchmark sourced from LeetCode, AtCoder, and CodeForces. Each problem is tagged by release date, so official evaluations can be restricted to post-cutoff time windows for contamination-aware comparison [18]. In the official code generation setting, the model receives the natural-language problem statement together with example input-output pairs and must return a complete executable program. Correctness is determined purely by functional correctness on unseen tests, using the benchmark harness rather than an LLM judge [18]. The benchmark paper reports *Pass@1* as the primary metric and also reports difficulty-stratified *Pass@1* using the platform-provided Easy/Medium/Hard labels. The public generation leaderboard [24] follows this same framing, and Table 1(b) therefore reports overall and difficulty-wise Pass@1 from the same leaderboard view. Our grading uses the official `lcb_runner` execution harness with a 10-second per-test-case timeout.

**BabyVision.** We use the BabyVision dataset release `UnipatAI/BabyVision` [36], which contains 388 image-question pairs spanning four task families: Fine-grained Discrimination (163), Visual Tracking (83), Spatial Perception (91), and Visual Pattern Recognition (51) [7]. The benchmark contains 135 multiple-choice items and 253 fill-in-the-blank items; in the official inference setup, each model is queried with a unified prompt template of the form "`{question} Think about the question and give your final answer in {format}.`", which standardizes answer formatting for downstream extraction and judging [7]. Official scoring uses an LLM-as-judge pipeline: given a model answer and the gold answer, Qwen3-Max determines semantic equivalence under the benchmark's fixed judge prompt, which the authors report as having perfect agreement with their human evaluators on the benchmark responses they audited [7]. The primary metric is *Avg@3*, defined in the paper as the mean *Pass@1* accuracy across three random runs, and the official tables additionally report the corresponding standard deviation; Table 1(c) follows the official overall-plus-family breakdown but keeps only the mean scores in the main text table. Our grading uses an LLM-as-judge pipeline with Claude Haiku 4.5 as the judge model, following the same semantic equivalence protocol as the official evaluation but substituting the judge model.

**AIME 2025.** AIME 2025 is treated as a final-answer mathematics benchmark on the MathArena competition table, which lists 30 problems for the 2025 competition and notes that each answer is an integer between 0 and 999 inclusive [27]. The task is constrained final-answer prediction: the model receives the official problem statement and is expected to return one final integer answer in the AIME answer space. Official grading is exact-answer grading against the released answer key, with no LLM judge and no partial credit, so each problem contributes a binary correct/incorrect label. The public leaderboard metric is *Accuracy*, namely the percentage of the 30 official problems answered exactly correctly; Table 4(a) reports that same public leaderboard metric. Our grading uses exact integer matching with no LLM judge, consistent with the official protocol.

**AIME 2026.** AIME 2026 appears on MathArena as another 30-problem final-answer competition, again with the official note that each answer is an integer between 0 and 999 inclusive [28]. The official task semantics match AIME 2025: the model is given the problem statement and must return one final integer answer. The benchmark performs exact-key grading with no LLM judge and no partial credit, and the leaderboard metric is therefore plain *Accuracy* over the 30 released competition problems. Keeping the same final-answer protocol across 2025 and 2026 allows direct year-to-year comparison under the benchmark's official grading rule; the corresponding public references are shown in Table 4(b). Our grading uses exact integer matching, consistent with the official protocol.

**GPQA-Diamond.** We use the GPQA release `Idavidrein/gpqa` [16], configuration `gpqa_diamond`, which corresponds to the 198-question Diamond subset introduced in the GPQA paper [31]. GPQA is a graduate-level four-option multiple-choice science benchmark spanning biology, physics, and chemistry, and GPQA-Diamond is the highest-stringency subset retained after expert-accuracy and non-expert-difficulty filtering [31]. The official task is closed-book answer selection unless otherwise stated: the model is shown the question and four candidate options and must output exactly one final option. Scoring is exact option-key accuracy with no semantic judge and

a random-guess baseline of $25\%$ [31]. In addition to Diamond-set accuracy, the original GPQA paper also reports accuracies on the Extended and Main sets and distinguishes closed-book few-shot/CoT baselines from an open-book GPT-4-with-search baseline; Table 1(d) keeps the main-text comparison focused on the Diamond-set accuracy column used by current public references. Our grading extracts the option letter (A–D) from the model response via regex and performs exact matching against the gold option key, with no LLM judge, consistent with the official protocol.

# NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction claim that ATTS matches or outperforms baselines on four main benchmarks while remaining cost-competitive; these claims are directly supported by Table 1 and the cost-accuracy analysis in Section 4.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 discusses limitations including shared misconceptions across candidates, lack of formal stopping guarantees, conservative early stopping behavior, increased latency on easy problems, and scope limited to reasoning benchmarks.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Appendix D provides two theorems (replication principle, information loss degrades synthesis) with complete proofs and explicitly stated assumptions. Both are specializations of Blackwell's informativeness theorem. Machine-verified proofs in Lean 4 are included in the code release.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 4.1 specifies the backbone model, sampling parameters, thinking budget, and exploration budget. Appendix A provides full implementation details including prompt templates, schemas, baseline implementations, and grading procedures. Appendix C documents all benchmark datasets, splits, and evaluation protocols.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Code will be released at `https://github.com/t2ance/ATTS`. All benchmarks used are publicly available with dataset identifiers cited in Appendix C. Reproduction requires access to the Claude API, which is a commercial service.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

   Answer: [Yes]

   Justification: No training is involved; this is a test-time inference method. Section 4.1 specifies the model, sampling parameters, thinking budget (32K tokens), and exploration budgets ($T$=8 single-model, $T$=16 multi-model). Appendix A details prompt design, grading procedures, and all baseline configurations. Appendix C specifies dataset splits and evaluation protocols for all six benchmarks.

7. **Experiment Statistical Significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: We report a 5-run stability analysis on GPQA-Diamond ($80.20\% \pm 0.55\%$), showing that orchestrator decisions are robust to sampling stochasticity. Other benchmarks are single-run; the GPQA analysis suggests reported numbers are representative.

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: All experiments use the Claude API (no local GPU training). Per-question costs in USD are reported for every method and benchmark in Tables 1–4, enabling direct cost estimation for reproduction.

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Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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Justification: The paper does not include a dedicated broader impacts discussion. The work is a general-purpose test-time scaling method for LLM reasoning and does not introduce domain-specific risks beyond those inherent to the underlying language models.

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper does not release pretrained models, scraped datasets, or other high-risk assets. The released code is an inference-time orchestration framework that calls existing commercial APIs.

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Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

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Justification: All benchmarks are cited with their original papers and HuggingFace dataset identifiers in Appendix C. The reward models (Skywork-Reward-V2, VisualPRM) are cited with both their papers and HuggingFace model cards. Claude models are accessed through the Anthropic API under its terms of service.

13. **New Assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

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